

ISHRAQ TASHDID

tashdid.ishraq@gmail.com — ishraq.tashdid@ucf.edu — +1 (689) 286-7719
Orlando, FL 32817 — [LinkedIn](#) — [GitHub](#) — [Google Scholar](#) — [ishraqtashdid.com](#)

Research Interests

Automated vulnerability reasoning and threat modeling (CWE/CVE/CAPEC); LLM-assisted security analysis across hardware–firmware–software boundaries; formal and property-based security verification; chiplet and SiP authentication; secure SoC design and supply chain integrity.

Education

University of Central Florida (UCF) *Jan. 2024–Present*
Ph.D. in Computer Engineering (Joint PhD/MS) GPA: **3.96**/4.00

Advisor: Dr. Sazadur Rahman, Department of Electrical & Computer Engineering

Trustee’s Doctoral Fellow (4-year, \$25,000/year) **Future Faculty Laureate**, UCF ECE

Bangladesh University of Engineering and Technology (BUET) *Feb. 2017–Jun. 2022*
B.Sc. in Electrical and Electronic Engineering CGPA: 3.48/4.00

Research Experience

Graduate Research Assistant — University of Central Florida *Jan. 2024–Present*
Hardware Security & Trustworthy Computing Lab, advised by Dr. Sazadur Rahman

- **LLM-driven threat modeling & automated vulnerability reasoning (DAC 2026):** Built *ATLAS*, a framework that ingests CWE/CVE/CAPEC vulnerability databases, auto-generates structured threat models for any SoC asset, and translates them into formal security properties verified by Jasper-Gold. Detected 39/48 known CWEs across three HACK@DAC benchmarks (>82% correct property generation). Conducted in collaboration with **Jonathan Valamehr (Intel Research)**, whose co-authorship reflects Intel’s institutional validation of the framework. Subsequently invited to present *ATLAS* to Intel’s AI research community at the **Intel AI Forum (Feb. 2026)**. Framework is directly applicable to firmware and software vulnerability workflows that reason over the same threat taxonomies.
- **Automated security property generation & cross-abstraction analysis:** Developed pipelines combining static analysis, AST-level structural reasoning, execution traces, and formal verification feedback to generate precise security assertions with minimal human effort, reducing reliance on manually crafted security rules at every abstraction layer from RTL through system firmware. This work forms the analytical backbone of both *ATLAS* and subsequent cross-abstraction security projects.
- **Chiplet authentication via PUF and MPC (HOST 2026):** Designed *InterPUF*, a distributed authentication framework for reconfigurable System-in-Package (SiP) interposers. Embedded a differential delay PUF in the interconnect fabric and secured evaluation using multi-party computation (MPC), ensuring raw PUF signatures never leave the device. Achieved <0.23% area and <0.072% power overhead; ML modeling attacks converged at ~47% accuracy (random-guess level), demonstrating resistance to state-of-the-art ML-based reverse engineering. Addresses supply chain integrity concerns in multi-vendor chiplet ecosystems flagged by the CHIPS and Science Act (2022) and DoD supply chain security reports.
- **Routing-based PUF authentication (GOMACTech 2026):** Designed *RoutePUF*, a routing-based physically unclonable function for secure interposer authentication in chiplet systems, extending the PUF-based authentication line of work to a government/defense-microelectronics audience via the Government Microcircuit Applications and Critical Technology Conference, a forum jointly sponsored by the U.S. Department of Defense.
- **Adaptive logic obfuscation for SoC IP protection (ICCD 2025):** Developed *ECOLoGic*, a framework enabling circular, obfuscated, and adaptive logic via eFPGA-augmented SoCs. The approach allows post-fabrication logic reconfiguration to resist reverse engineering and hardware Trojan insertion while maintaining functional correctness and low overhead.
- **Macro placement automation (MLCAD 2025 — Best Paper Nominee):** Designed human-inspired reinforcement learning frameworks for chip floorplanning to improve post-route routability and reliability in heterogeneous SoCs. Demonstrated that incorporating human design intuition into RL

reward modeling yields measurable gains over PPA-only optimization baselines.

- **Scalable chiplet trust architectures:** Developed *SAFE-SiP* (GLSVLSI 2025) and *AuthenTree* (PAINE 2025), two complementary MPC-based distributed trust frameworks addressing authentication scalability in chiplet-based heterogeneous integration. AuthenTree achieves logarithmic communication complexity in the number of chiplets, enabling practical deployment in large-scale SiP designs.

Industry Experience

XPU Security Research Intern — Intel Corporation, Folsom, CA

May 2026–Present

- Competitive research internship in Intel’s XPU (accelerator) security division. Selected following an invited talk at the Intel AI Forum on LLM-driven threat modeling for SoC security (Feb. 2026). Focused on security research for next-generation accelerator architectures, working with Intel’s AI and hardware security research community. Supervisor: James Cavanaugh, Intel.

Hardware Verification Engineer — XCelerium, Irvine, CA (via DSI, Bangladesh) *Jan. 2022–Dec. 2023*

- Verified custom **RISC-V** processor implementations at the ISA and **system level** using SystemVerilog, UVM, and cocotb, developing deep familiarity with the hardware–software boundary where firmware executes and privilege/mode transitions occur.
- Verified **AXI memory controllers** and **APB-AXI bridges**, including CDC edge-cases and assertion-based checks on bus-level access control — the same protocol surfaces targeted by firmware and software security assessments.
- Built **Python-driven automation frameworks** for coverage convergence, testbench orchestration, and results analysis, accelerating sign-off across multiple tapeout projects.

Publications

- **Ishraq Tashdid**, Kimia Tasnia, Alexander Garcia, Jonathan Valamehr, Sazadur Rahman, “*ATLAS: AI-Assisted Threat-to-Assertion Learning for System-on-Chip Security Verification*,” **DAC 2026** (63rd Design Automation Conference), Long Beach, CA, USA. *Accepted*. [arXiv:2603.01170](#)
- **Ishraq Tashdid**, “*RoutePUF: A Routing-Based Physically Unclonable Function for Secure Interposer Authentication in Chiplet Systems*,” **GOMACTech 2026** (Government Microcircuit Applications and Critical Technology Conference). *Accepted*.
- **Ishraq Tashdid**, Tasnuva Farheen, Sazadur Rahman, “*InterPUF: Distributed Authentication via Physically Unclonable Functions and Multi-party Computation for Reconfigurable Interposers*,” **HOST 2026** (IEEE Int’l Symposium on Hardware Oriented Security and Trust), Washington, D.C., USA. *Accepted*. [arXiv:2601.11368](#)
- **Ishraq Tashdid**, Dewan Saiham, Nafisa Anjum, Tasnuva Farheen, Sazadur Rahman, “*ECOLogic: Enabling Circular, Obfuscated, and Adaptive Logic via eFPGA-Augmented SoCs*,” **ICCD 2025** (IEEE Int’l Conference on Computer Design), Dallas, TX, USA. [arXiv:2508.04516](#)
- **Ishraq Tashdid**, Tasnuva Farheen, Sazadur Rahman, “*SAFE-SiP: Secure Authentication Framework for System-in-Package Using Multi-party Computation*,” **GLSVLSI 2025** (ACM Great Lakes Symposium on VLSI), New Orleans, LA, USA. [arXiv:2505.09002](#)
- **Ishraq Tashdid**, Tasnuva Farheen, Sazadur Rahman, “*AuthenTree: A Scalable MPC-Based Distributed Trust Architecture for Chiplet-based Heterogeneous Systems*,” **PAINE 2025** (IEEE Physical Assurance and Inspection of Electronics), Denver, CO, USA. [arXiv:2508.13033](#)
- **Ishraq Tashdid**, Valentina Terry, Jordan Merkel, Tasnuva Farheen, Sazadur Rahman, “*BeyondPPA: Human-Inspired Reinforcement Learning for Post-Route Reliability-Aware Macro Placement*,” **ML-CAD 2025** (ACM/IEEE Symposium on Machine Learning for CAD), Santa Cruz, CA, USA. **Best Paper Nominee**. [DOI:10.1109/MLCAD65511.2025.11189164](#)
- **Book Chapter (invited): LLMs for SoC Design and Security** — with collaborators at the University of Florida, *in preparation*, 2025–2026.

Invited Talks & Presentations

Intel AI Forum — Intel Corporation, Folsom, CA

Feb. 2026

Invited (non-submitted) talk on *ATLAS*: LLM-driven threat modeling and formal security verification for SoC security. Presented to Intel’s AI research community of engineers and researchers; selected by Intel

organizers from ongoing academic collaborations. Directly led to selection for Intel’s competitive XPU Security Research Internship.

4th Annual Florida Semiconductor Summit (FSI) — Rosen Shingle Creek, Orlando, FL *Feb. 2026*
Invited presentation on hardware security and AI-assisted verification for Florida’s semiconductor manufacturing ecosystem. Theme: *Semiconductor Manufacturing in Florida: Power. Progress. Possibilities.* Organized by Florida Semiconductor Institute.

Open-Source EDA Birds-of-a-Feather — 63rd Design Automation Conference (DAC 2026), Long Beach Convention Center, Long Beach, CA *Jul. 2026*
Invited rapid-fire presentation, “*Security Challenges and Opportunities in the Open-Source EDA Ecosystem,*” at the Open-Source EDA, Data and Collaboration Summit — a recurring DAC Birds-of-a-Feather forum (running since 2018) organized by UC San Diego, Google, and Precision Innovations, bringing together academic researchers, industry practitioners, and open-source EDA maintainers.

Teaching & Mentoring

Graduate Teaching Assistant — UCF ECE *Spring 2024; Summer & Fall 2025*

- Courses: Engineering Analysis and Computation; Linear Systems II; Electronics II Lab; Digital Systems Lab. Responsibilities included office hours, grading, and lab instruction.

Undergraduate Research Mentor *Aug. 2024–Present*

- Mentored two undergraduate researchers in the UCF hardware security lab. Both students secured **summer 2025 research internships at AMD and Northrop Grumman**, respectively, and co-authored the **MLCAD 2025 Best Paper Nominee**. Northrop Grumman placement reflects direct contribution to the US defense contractor engineering pipeline.

Professional Service

Program Committee, AAAI-26 Fall Symposium Series (FSS-26)

External Reviewer, Design Automation Conference (DAC 2026 and DAC 2025)

External Reviewer, IEEE International Conference on Computer Design (ICCD 2025)

External Reviewer, IEEE International Symposium on Hardware Oriented Security and Trust (HOST 2026)

External Reviewer, IEEE Design & Test (journal)

Serving on a program committee and invited by program committees and editorial boards to review submissions spanning EDA, hardware security, computer design, and artificial intelligence — across flagship conferences, an AAAI symposium series, and an archival IEEE transactions-family journal. These invitations reflect recognition by field leaders as a qualified independent expert.

Patents

Provisional Patent Application *2026*

Filed on the *ATLAS* LLM-driven threat-modeling and automated security-property generation framework (DAC 2026), with Dr. Sazadur Rahman and Kimia Tasnia. (*Provisional/preliminary stage.*)

Grants & Proposals

Research Contributor *2026*

DOE Genesis Mission: Transforming Science and Energy with AI (RFA DE-FOA-0003612), PI: Dr. Sazadur Rahman, in collaboration with **Silvaco Inc.** (*Submitted; pending review.*)

Contributed to proposal development and technical research direction under DOE’s \$293M national AI initiative, targeting the semiconductors and microelectronics challenge area. Technical content builds directly on *ATLAS* and related security-verification publications. Reflects early-career engagement at the grant-writing and research leadership level in a White House-led federal program.

Research Contributor *2025*

ASCEND — NSF Secure and Trustworthy Cyberspace (SaTC 2.0) Program, PI: Dr. Sazadur Rahman. (*Submitted; pending review.*)

Contributed to the proposal’s technical research direction on security and trust for next-generation heterogeneous chiplet integration and advanced packaging, extending authentication and trust-architecture methodology from InterPUF, SAFE-SiP, and AuthenTree.

Research Contributor2025

SecureOpenROAD — NSF Safety, Security, and Privacy in Open Source Ecosystems (Safe-OSE, NSF 24-608), PI: Dr. Sazadur Rahman, in collaboration with the OpenROAD Initiative (ORI). *(Submitted.)*
Contributed to the proposal’s technical research direction on strengthening security assurance within the OpenROAD open-source chip-implementation ecosystem, building on research first presented at the DAC 2026 Open-Source EDA Birds-of-a-Feather session.

Honors, Awards & Fellowships

- **UCF Trustees Doctoral Fellowship** (2024–2028) — 4-year fellowship (\$25,000/year) awarded to newly enrolled doctoral students with outstanding records. Among the most selective doctoral awards at UCF; recipients are identified for exceptional academic and research potential at the time of admission.
- **UCF ECE Outstanding Student Researcher Award** (May 2026) — Departmental recognition for outstanding contributions to research during FY 2025–26, awarded by the UCF Department of Electrical and Computer Engineering.
- **DAC Young Fellow** — 63rd Design Automation Conference (DAC 2026), Long Beach, CA. Competitive recognition selecting outstanding early-career researchers at the premier international conference on electronic design automation.
- **Future Faculty Laureate (FFL)** — UCF ECE Department (Fall 2024–Present)
Competitive program identifying Ph.D. students with exceptional potential for academic faculty careers. Participants receive mentorship, teaching development, and professional preparation for faculty positions at research universities.
- **Best Paper Award Nominee** — ACM/IEEE MLCAD 2025
Nominated by program committee among top papers at the Symposium on Machine Learning for CAD.
- **NSF Travel Grant** — IEEE International Symposium on Hardware Oriented Security and Trust (HOST 2025). Federal recognition of research merit; awarded competitively to support participation in the premier hardware security symposium.

Technical Skills

Security & Analysis:	Threat modeling (CWE/CVE/CAPEC), LLM-assisted vulnerability reasoning, assertion-based verification, formal verification (JasperGold), static analysis, hardware IP protection
Programming:	Python, SystemVerilog, Verilog, C, MATLAB, Assembly (RISC-V)
Verification:	UVM, cocotb, pytest, ABV; RISC-V ISA & system-level compliance; AXI/APB bus protocols
EDA & Simulation:	Cadence Innovus / Genus / JasperGold, ModelSim, Quartus Prime, Yosys, PyVerilog
ML / AI:	PyTorch, reinforcement learning (PPO/DDPG), LLM prompting & fine-tuning, RAG pipelines